Q1. Big Data is very large data that is difficult to manage.
It cannot be handled by normal software.
So special tools are used.

Q2. MapReduce splits data and processes it.
It has two steps: map and reduce.
It helps in handling large data.

Q3. Big Data has 5 V's like Volume, Velocity, and Variety.
They tell about data size and speed.
Also about types of data.

Q4. Big Data has three types.
They are structured and unstructured data.
Also semi-structured data.

Q5. Hadoop is used for storing big data.
It also helps in processing data.
It works on multiple systems.

Q6. **Introduction:**
HDFS is used to store big data in Hadoop. It works on many computers.

Architecture:
It has two main components: NameNode and DataNodes.
The NameNode controls the system, and DataNodes store the data.

Files are divided into blocks and stored in different nodes.

Working:
Client first contacts NameNode. Then it gets data from DataNodes.

Some important points:

- Data is split into blocks
- Blocks are stored in different systems
- Copies of data are created

This helps in managing large data easily.

Conclusion:
HDFS is useful for storing big data and works efficiently.

Q7. Replication means storing copies of data.
In HDFS, each block has multiple copies.

This is important because:

- Data is safe
- Failure does not affect access

If a DataNode fails, NameNode detects it using heartbeat.
Then it creates new copies of data.

Users can access data from other nodes.
So data is always available.